

Fair and Constraint-Aware Radiology Shift Scheduling: A Computational Perspective on Fairness, Robustness and Explainability

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Abstract

This project develops and evaluates a decision-support system for monthly shift scheduling of radiology technicians at the Hospital of Covilhã, Portugal. The real process is characterised by minimum coverage requirements, heterogeneous shift eligibility, vacations and other absences, working-time limits, rest rules, individual preferences, and fairness obligations that extend across months. The system formalises these requirements as a constraint optimisation problem and solves it with the CP-SAT solver. Hard constraints encode conditions that must not be violated; weighted soft constraints represent preferences and operational compromises. The evaluation addresses three dimensions of Trustworthy AI. Fairness is assessed through raw and availability-normalised workload, night and weekend burdens, Gini coefficients, and Jain indices, complemented by five objective-weight profiles. Robustness is evaluated in six controlled disruption scenarios, all of which produced feasible repairs while changing between 20 and 29 assignments. Explainability is supported through a published constraint catalogue, assignment-level rule facts, non-assignment diagnostics, repair explanations, and counterfactual re-solving. A September 2026 case study demonstrates strong equality in night allocation and enforces a hard minimum of six imaging workers on every weekday morning. Meeting this floor requires 316 overtime hours shared by 13 workers, while the preferred target of 10 remains unmet on 10 days. This result illustrates why optimisation status must not be confused with complete operational adequacy. The system is therefore presented as an auditable assistant for human schedulers, not an autonomous authority.

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1 Introduction

Monthly hospital scheduling is a high-impact allocation problem. A timetable distributes paid work, nights, weekends, inconvenient sequences, and opportunities for rest among employees, while also determining whether a clinical service has adequate staff. These outcomes are ethically relevant even when no sensitive attribute is used: apparently neutral rules can concentrate undesirable work, and a mathematically feasible solution can still encode an unacceptable operational compromise.

The problem becomes particularly difficult when ordinary contractual capacity is insufficient to meet demand. Overtime must then be allocated, which necessarily means that some workers will perform more hours than others. In a manual process, decisions about who receives this additional burden may be influenced—or may be perceived as being influenced—by personal relationships, inconsistent judgement, or the immediate pressure of producing a timetable. A constraint solver offers a useful form of procedural consistency: it can apply the same declared rules to every eligible worker, consider carried workload and availability, and record why unequal hours were required. Delegating the combinatorial allocation to a solver can therefore support a fairer process and reduce room for ad hoc favouritism. It does not, however, make the result automatically fair, because the solver still implements human choices about inputs, constraints, priorities, and objective weights.

This project adapts the course proposal on fair and constraint-aware shift scheduling to a real radiology department. Its objective is not only to produce a timetable, but to make the assumptions, compromises, and consequences of that timetable visible. The main research question is:

How can operational feasibility, worker fairness, resilience to disruption, and understandable justification be represented and evaluated in one scheduling decision-support system?

The work makes four contributions:

- a configurable constraint model that distinguishes mandatory rules from weighted preferences;
- worker-level and aggregate fairness evaluation, including normalisation by opportunity;
- controlled robustness tests that quantify repair feasibility and schedule instability;
- factual, local, repair, and counterfactual explanations with explicit epistemic limits.

The September 2026 configuration is the principal case study. It contains 21 workers, 30 calendar days, individual vacations, consultation assignments, sick leave, contract data, carried rest, and rotation histories. A smaller hemodynamics sub-roster is also supported. All personal names and operational data used here were supplied for the scheduling case study; the report does not infer protected characteristics or evaluate demographic groups.

2 Employed Technologies

2.1 Python and OR-Tools CP-SAT

The implementation is written in Python. The scheduling model uses Google OR-Tools CP-SAT, a constraint-programming solver for integer and Boolean models. CP-SAT is appropriate because shift assignments are discrete, many rules are logical or cardinality constraints, and the objective is a weighted sum of penalties. It can return a feasible solution under a time limit together with an objective bound; this distinction is retained in the generated solver report.

2.2 Data and reporting stack

Inputs and outputs use JSON and CSV so that assumptions and schedules can be inspected without specialised software. Matplotlib and NumPy generate timetable, fairness, trade-off, staffing, overtime, and robustness figures. The implementation is divided into the solver (`solver.py`), fairness evaluator (`evaluate.py`), experiment runner (`experiments.py`), robustness suite (`robustness.py`), explanation generator (`explanations.py`), and visual reporting utilities.

2.3 Reproducibility and testing

Automated tests cover calendar functions, constraints, validation, fairness metrics, experiment outputs, robustness scenarios, and explanations. Fixed configuration files make the evaluated month reproducible. Generated solver reports include status, solve time, objective value, best bound, weights, unweighted objective components, and output locations.

3 Scheduling System

3.1 Inputs and outputs

For a selected month, the system reads the roster, vacations, consultations, sick leave, contract hours, prior-month rest debt, workday history, holiday rotation history, and optional demand adjustments. Its primary output is a row-oriented schedule containing date, worker, and shift. It additionally produces compensatory-rest records, prevention duty, fairness metrics, figures, solver diagnostics, and carried state for the next month.

Four shift codes appear in the implementation: morning (M, 08:00–16:00), afternoon (A, 16:00–00:00), early afternoon (EA, 14:00–22:00), and night (N, 00:00–08:00). All have eight-hour duration. Eligibility differs by worker: for example, some workers do not work nights, one worker is morning-only, and one works weekday mornings or afternoons only.

3.2 Decision variables

Let P be the workers, D the dates in the month, and $S = \{M, A, N\}$ the main shift categories. The central Boolean variable is

$$x_{pds} = \begin{cases} 1 & \text{if worker } p \text{ is assigned to shift } s \text{ on day } d, \\ 0 & \text{otherwise.} \end{cases}$$

Additional Boolean and integer variables represent the early-afternoon variant, prevention duty, morning participation, daily morning shortfall, monthly overtime, workload deviations, patterns of consecutive work, and schedule changes during repair.

3.3 Hard constraints

The model catalogues each hard constraint so that the same language can be reused in reports and explanations. The principal rules are:

- at most one shift per worker and calendar day;
- availability and worker-specific shift eligibility;

- weekly working time not exceeding 40 hours;
- monthly contractual caps adjusted for carried rest, with explicitly bounded overtime when enabled;
- no morning immediately after a standard afternoon;
- no night immediately after an early afternoon;
- no work on the day after a night and minimum spacing between nights;
- floor/ceiling sharing of night shifts among eligible workers;
- minimum staffing for afternoons and nights, a hard weekday-morning floor of six, and maximum weekday-morning staffing;
- hemodynamics-specific prevention alternation, compensatory rest, and morning-presence rules.

The weekday morning requirement has two levels. Six workers are an unconditional hard minimum, while 10 workers remain the preferred operational target. Any gap between six and 10 is represented by a non-negative slack variable. On Monday, Wednesday, and Friday it receives a weight of 10,000,000; on Tuesday and Thursday it receives a weight of 500. This design permits transparent prioritisation under limited capacity without ever producing a weekday morning with fewer than six assigned imaging workers.

3.4 Soft preferences and objective

Let $c_k(x)$ be the violation count or magnitude for preference k and w_k its configured weight. The solver minimises

$$\min_x \sum_k w_k c_k(x).$$

The terms cover cumulative workday balance, proportional workload deviation, long work streaks, isolated work or rest days, long afternoon sequences, preparation patterns before nights, nights immediately after vacation, holiday rotation, weekend imbalance and continuity, vacation-adjacent weekends, schedule preservation, morning deficits, and overtime.

Weights are not moral truths. They encode an operational prioritisation and can make unlike penalties commensurable only by design choice. For this reason the system saves both raw and weighted components and compares multiple profiles rather than treating one objective as uniquely correct.

3.5 Solution workflow

The workflow has five stages. First, the configuration is validated and converted to calendar-indexed sets. Second, the constraint model is built. Third, CP-SAT searches for a schedule under a time limit. Fourth, the solution is independently summarised through coverage, fairness, and carried-state reports. Finally, visualisations and explanations are generated for human review. A repaired schedule uses the published schedule as a reference and adds a high penalty for assignment changes.

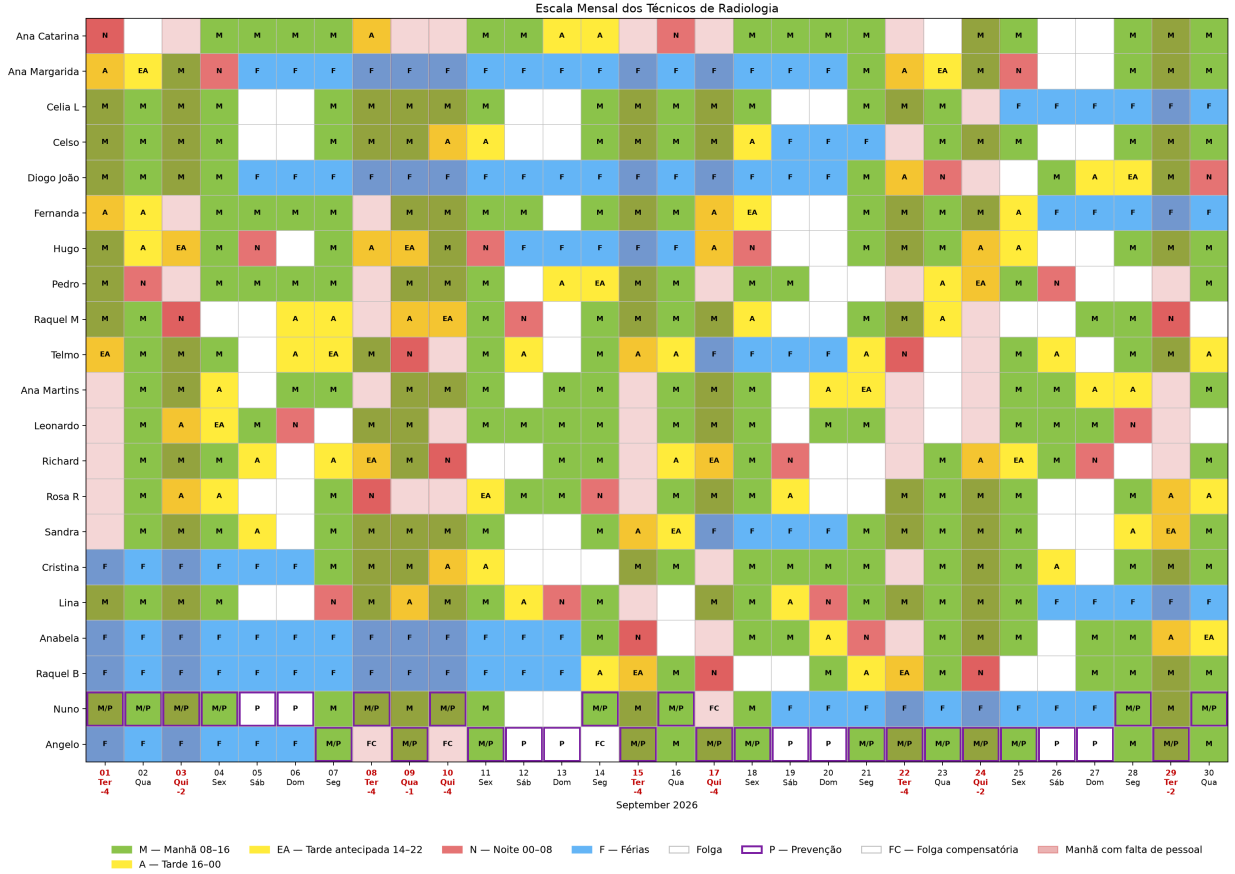


Figure 1: Generated September 2026 timetable. The visual is an operational review aid; the CSV and solver report remain the authoritative machine-readable outputs.

4 Trustworthy-AI Analysis

4.1 Fairness

4.1.1 Fairness targets

The system considers several distributive goods and burdens: total assigned hours, workload relative to days of availability, nights, nights relative to night opportunities, weekends worked, and weekends relative to available weekends. Opportunity normalisation matters because workers have different vacations and eligibility. Equal raw night counts, for example, are neither possible nor meaningful when some workers cannot work nights.

For a non-negative vector $y = (y_1, \dots, y_n)$, inequality is measured with the Gini coefficient

$$G(y) = \frac{\sum_i \sum_j |y_i - y_j|}{2n \sum_i y_i},$$

where 0 denotes equality. Jain's index is

$$J(y) = \frac{(\sum_i y_i)^2}{n \sum_i y_i^2},$$

where 1 denotes equality. Both are descriptive summaries, not proofs of justice. Raw-hours metrics include workers unavailable for the entire month; eligibility-normalised night and weekend metrics exclude workers with no opportunity for that burden.

4.1.2 Group fairness and data minimisation

The project proposal anticipated group comparisons where suitable data were ethically available. No reliable protected-attribute dataset is present in the implementation, and attributes must not be inferred from names. Consequently, this report does not claim demographic parity or absence of discrimination. It instead evaluates individual and eligibility-conditioned distributions. This is a limitation, but also an application of data minimisation: sensitive data should be collected only with a defined governance purpose, lawful basis, and appropriate safeguards.

4.2 Technical robustness and safety

Robustness is defined here as the capacity to repair a published schedule after a bounded disruption while retaining legal feasibility and limiting changes imposed on workers. Six scenarios cover one-day absence, absence of a scheduled night worker, three-day absence, simultaneous absences, and increased morning or night demand. For each scenario the suite records status, time, assignment edits, affected workers, shortfall, and fairness deltas.

Safety is not equivalent to solver feasibility. The application therefore reports coverage slack explicitly and preserves the original schedule during repair where possible. A human scheduler remains responsible for deciding whether a shortfall is operationally tolerable, whether agency or cross-unit staff are available, and whether a repaired timetable can be published.

4.3 Explainability

The explanation design separates three questions:

1. *What rules govern the system?* A global catalogue describes constraints, preferences, weights, and solver guarantees.
2. *What facts make an assignment legal and operationally relevant?* Local reports state coverage, eligibility, availability, weekly/monthly hours, night spacing, and special-shift facts.
3. *What would happen if an assignment changed?* Counterfactual re-solving forbids a selected assignment and reports whether an alternative schedule exists and what changes.

The local reports deliberately state that legality and tight coverage do not prove a unique cause. In a global optimisation model, an assignment may be one of many equivalent choices. Counterfactual solving provides stronger evidence about alternatives, although the result still depends on the encoded constraints, weights, time limit, and input accuracy.

5 Experimental Evaluation

5.1 Case-study setup

The main experiment schedules September 2026. The configuration contains 21 radiology workers; 13 are night-eligible and 18 have at least one weekend opportunity. Cátia is on sick leave for the entire month and Ana Sofia is assigned to consultations throughout the month, so both receive zero rostered shifts. Other workers have heterogeneous vacation periods. These facts materially affect raw workload equality.

The final principal solver run used a 60-second limit. CP-SAT returned FEASIBLE, with objective 16,326,175, best bound 16,318,680, and a relative gap of approximately 0.046%. This is strong

but not certified optimality; the report therefore uses “feasible schedule” rather than “optimal schedule.”

5.2 Operational result

Most active and available full-time workers receive 20 shifts (160 hours). Workers with long absences receive 12 or 13 shifts, while Lina receives eight imaging shifts because her hemodynamics work is accounted for separately. Enforcing the six-person weekday-morning floor requires 316 overtime hours: 12 workers receive 24 extra hours and Lina receives 28. No excess-work-streak, excess-afternoon-streak, or night-after-vacation penalties occur.

No weekday morning has fewer than six imaging workers. Ten dates remain below the preferred target of 10, totalling 31 missing worker-shifts relative to that target: September 1, 3, 8, 9, 10, 15, 17, 22, 24, and 29. Actual coverage on these dates ranges from six to nine. The result shows that the hard safety floor can be met, but only through extensive overtime, and that full target coverage still conflicts with availability and monthly capacity. Deployment would therefore require explicit approval of the overtime burden and consideration of additional staffing or cross-coverage.

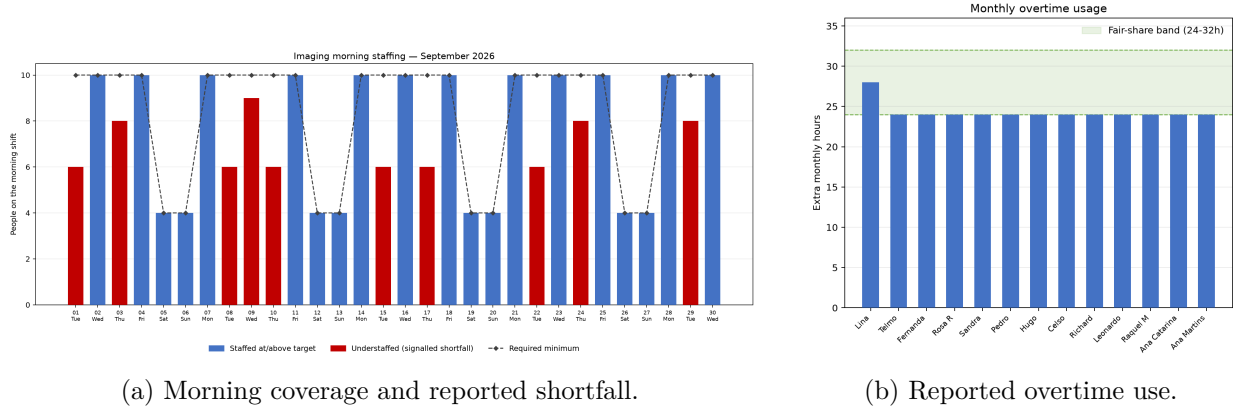


Figure 2: Capacity diagnostics for the September case study.

5.3 Fairness results

Table 1 reports the main fairness results. Night allocation is the most even distribution: eligible workers receive two or three nights, with $G = 0.0923$ and $J = 0.9615$. Raw assigned hours appear less equal because two workers have no roster opportunity and several have long absences. Availability normalisation improves interpretability but does not eliminate inequality. Weekend burden is less even than night burden, particularly after normalisation.

Table 1: September 2026 fairness metrics.

Metric	Mean	Min.	Max.	Gini	Jain
Assigned hours	126.86	0	160	0.1876	0.8652
Availability-normalised workload	0.6562	0	0.9524	0.1712	0.8789
Night shifts (eligible workers)	2.3077	2	3	0.0923	0.9615
Normalised night burden	0.1004	0.0667	0.1429	0.1545	0.9286
Weekends (eligible workers)	2.2222	0	4	0.3194	0.7533
Normalised weekend burden	0.7037	0	1	0.2522	0.8155

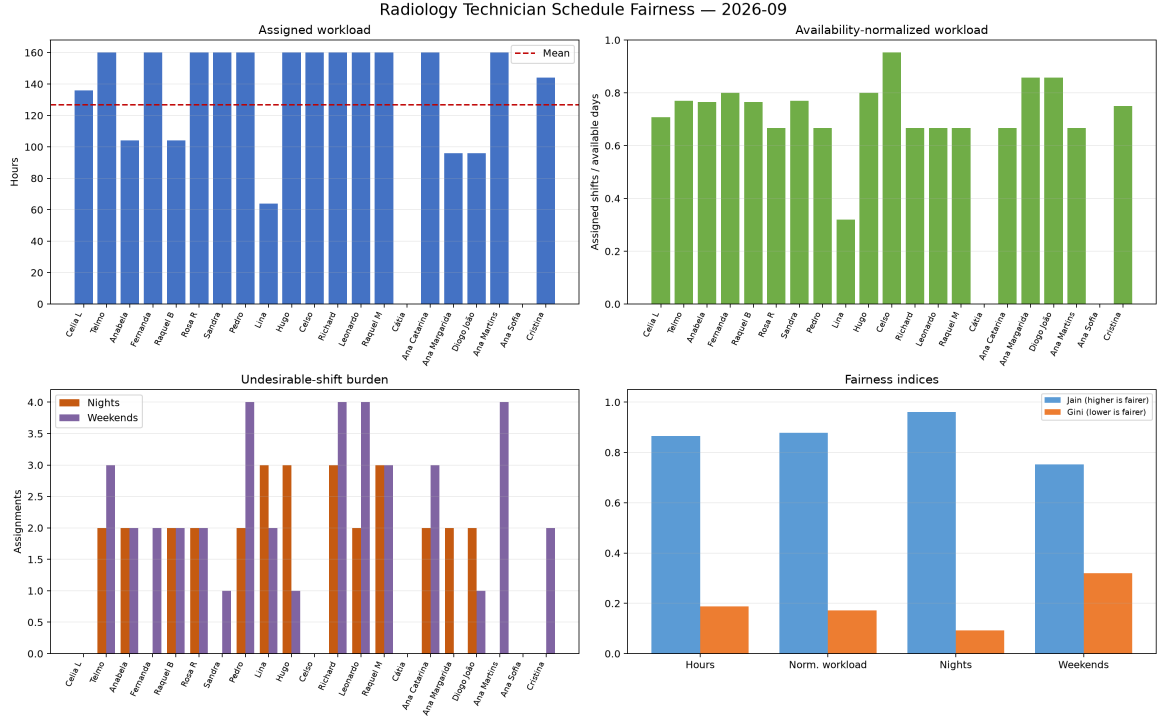


Figure 3: Worker-level workload and undesirable-shift burdens, with aggregate indices.

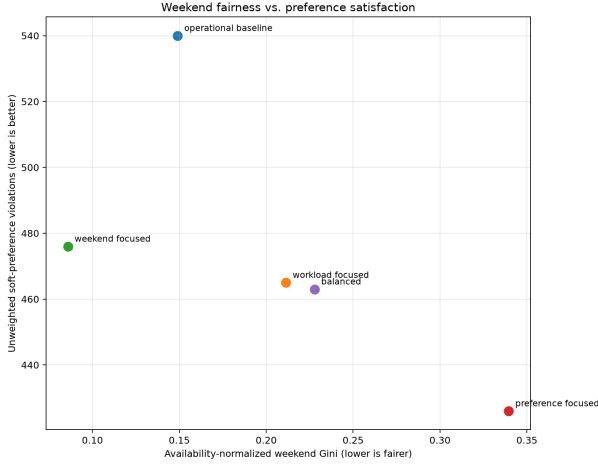
5.4 Trade-off experiments

Five objective profiles were evaluated: operational baseline, workload-focused, weekend-focused, preference-focused, and balanced. The profiles are not directly ranked by their raw objective values because their weights differ. Instead, Table 2 compares common output metrics.

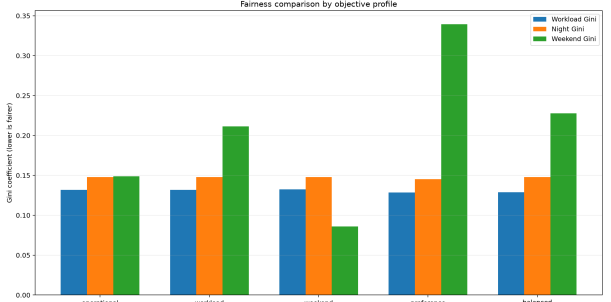
The preference-focused profile has the best normalised-workload Gini (0.1286) and lowest reported preference cost (426), but the worst weekend Gini (0.3395). The weekend-focused profile achieves the best weekend Gini (0.0859), at a preference cost of 476. Raw workload Gini varies only from 0.1780 to 0.1836 because the hard coverage floor and required overtime strongly constrain total workload. The profiles therefore mainly redistribute weekend and pattern burdens rather than eliminating the underlying capacity requirement.

Table 2: Comparison of objective profiles. “Pref.” is the unweighted aggregate preference cost produced by the experiment runner.

Profile	Status	Workload G	Norm. work G	Night G	Weekend G	Pref.
Operational baseline	Optimal	0.1780	0.1317	0.1480	0.1490	540
Workload-focused	Feasible	0.1780	0.1317	0.1480	0.2114	465
Weekend-focused	Feasible	0.1809	0.1326	0.1480	0.0859	476
Preference-focused	Feasible	0.1824	0.1286	0.1454	0.3395	426
Balanced	Feasible	0.1836	0.1287	0.1480	0.2278	463



(a) Fairness–preference trade-off.



(b) Fairness indices across profiles.

Figure 4: Objective weights select different distributive outcomes; no profile dominates on every measure.

5.5 Robustness results

All six disruption scenarios produced feasible, solver-certified optimal repairs in less than one second. The repairs changed 20 to 29 assignment rows, corresponding to 10 to 15 replacement-equivalent changes, and affected nine or 10 workers. The stronger coverage floor consequently makes repairs less local than in the earlier formulation. Workload Gini deltas lie between -0.0403 and -0.0378 ; night Gini deltas between -0.0168 and 0 ; weekend Gini deltas between -0.0265 and -0.0065 .

These results support feasibility under the tested perturbations, but also show that repairs can propagate across much of the roster when the system is operating close to its capacity floor. They do not establish resilience to larger outbreaks, extended understaffing, incorrect contract data, or a change in the legal interpretation of rest rules. “Staffing shortfall” in this experiment measures distance from the preferred target, not violation of the six-person hard floor.

Table 3: Robustness scenarios and schedule disruption.

Scenario	Edits	Replacements	Workers	ΔG_W	ΔG_N
Single-day absence	23	12	9	-0.0397	-0.0140
Night-worker absence	23	12	9	-0.0397	-0.0011
Three-day absence	22	11	9	-0.0399	-0.0168
Morning-demand increase	20	10	10	-0.0378	0.0000
Night-demand increase	21	11	9	-0.0393	-0.0127
Simultaneous absences	29	15	10	-0.0403	-0.0140

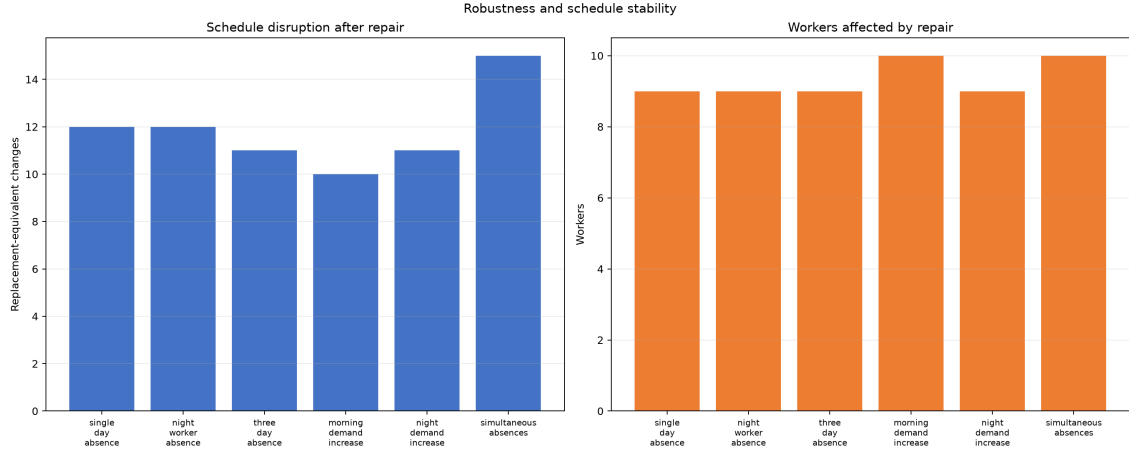


Figure 5: Assignment changes and fairness effects under controlled disruptions.

5.6 Explanation examples

Assignment explanations verify that a worker was available and eligible, show achieved coverage, place the assignment in weekly and monthly hour totals, and report relevant night-spacing or early-afternoon facts. Non-assignment explanations identify blocking rules or capacity limits. Repair explanations list added and removed assignments after a disruption. Counterfactual explanations rerun the solver with a selected assignment forbidden and compare the alternative timetable.

This layered design is preferable to a single natural-language rationale because it distinguishes verified facts from optimisation inference. It also supports contestability: a worker or scheduler can challenge an input fact, a rule, a weight, or the practical acceptability of the resulting trade-off.

6 Ethical Discussion

6.1 What “fair” means in this system

The project operationalises fairness as a family of inspectable distributions rather than a single score. Equality of raw workload rewards equal totals; equity relative to availability adjusts for opportunity; burden sharing focuses on nights and weekends; temporal fairness carries workday and holiday history across months. These principles can conflict. Equal hours may overload a worker with limited availability, while equal opportunity-normalised ratios may create unequal pay or totals. The correct balance requires institutional and worker participation, not only optimisation.

Overtime makes this distinction concrete. When staffing demand cannot be met within ordinary contractual hours, perfect equality may be impossible: one or more workers must accept additional hours, and operational restrictions may make only a subset eligible. The relevant fairness question is then not whether every worker receives an identical total, but whether the unequal burden is necessary, proportionate, transparently calculated, and fairly rotated over time. The model represents overtime explicitly, penalises its use, bounds the amount assigned to an overtime worker, and reports the resulting allocation. This is preferable to leaving the choice implicit in a manually assembled roster.

Using a solver can also improve procedural fairness. Unlike an individual scheduler, the optimisation engine has no friendships, grudges, or personal preference for particular employees; given the same model, data, and solver conditions, it applies the encoded criteria consistently. This can reduce opportunities for conscious or unconscious favouritism and makes decisions easier to audit.

Nevertheless, neutrality of the computation is not neutrality of the whole system. Historical records may contain earlier inequities, input data may be incomplete, and weights may privilege some interests over others. Human oversight is therefore needed to define legitimate rules, inspect unequal outcomes, and correct the model rather than to make unexplained exceptions for favoured individuals.

The night-fairness band is a relatively strong distributive commitment because it constrains eligible workers to floor/ceiling shares. Other fairness goals are weighted and can be traded against pattern preferences or staffing. The reports make that distinction visible.

6.2 Human oversight and contestability

The intended workflow is human-in-the-loop. Before publication, a scheduler should verify inputs, coverage, overtime, compensatory rest, and unusual individual burdens. Workers should have a route to correct availability or contract data and contest undesirable patterns. After approval, any automated repair should be reviewed and communicated to affected staff. The generated explanations assist these tasks but do not replace consultation.

6.3 Privacy and data governance

The schedule contains personal and potentially sensitive employment information. Access should be restricted to authorised users; generated reports should be stored and transmitted securely; public or classroom artefacts should be pseudonymised where possible. Retention periods should be defined, especially for sickness, vacation, and historical burden data. Additional demographic data should not be collected merely to improve a dashboard; group-audit data require a clear purpose and governance process.

6.4 Failure modes

Important failure modes include incomplete absence data, mistaken shift eligibility, outdated contractual hours, incorrect carry-over state, overconfident interpretation of a feasible status, and weights that silently privilege one group or pattern. A second class of failures arises from model omission: skill mix, ward-specific competence, emergency escalation, and qualitative team considerations may not be fully represented. Finally, an explanation may be technically true yet misleading if it describes legality without mentioning a service-wide capacity shortfall. The reporting design mitigates, but cannot eliminate, these risks.

7 Limitations

The evaluation concerns one main month and a bounded set of synthetic disruptions. It does not constitute a prospective clinical deployment study. There is no comparison against historical manual schedules or worker satisfaction data, so claims about improved fairness or administrative time cannot yet be made. The objective weights have not been elicited through a formal multi-stakeholder process. Group fairness is not assessed because protected attributes are unavailable. Statistical metrics are descriptive and sensitive to the included population and opportunity definitions.

The weekday-morning formulation now prevents coverage below six, while retaining 10 as the preferred target. A production interface should display both hard-floor compliance and distance from the operational target. It should also require explicit approval of the substantial overtime needed to maintain the floor.

Counterfactual explanations may be computationally expensive and can vary with the solver time limit. Local explanations do not yield a complete causal decomposition of a global optimum. Finally, the code relies on the accuracy of local configuration files and has not been independently audited for compliance with Portuguese employment law or hospital policy.

8 Conclusions and Future Work

The project demonstrates that radiology scheduling can be represented as an auditable optimisation problem without reducing Trustworthy AI to a single fairness score. The implementation generates reproducible schedules, exposes hard rules and weighted compromises, measures distributional outcomes, repairs bounded disruptions with few changes, and produces explanations that acknowledge their limits.

The September case study has two central findings. First, night work can be distributed quite evenly among eligible workers: the night-count Gini is 0.0923 and Jain index is 0.9615. Second, a hard floor of six weekday-morning workers is feasible, but it requires 316 overtime hours across 13 workers and still leaves 10 days below the preferred target of 10. This is precisely the type of ethically and operationally important trade-off that a transparent system should reveal rather than conceal.

Future work should validate every constraint and weight with schedulers and worker representatives, compare generated and historical rosters, add formal skill-mix requirements, distinguish critical from negotiable coverage in the user interface, and evaluate worker-perceived fairness. Longer multi-month experiments should test carried fairness and holiday rotation. A deployment study should measure correction rates, planning time, stability, contestation outcomes, and whether explanations improve appropriate trust rather than simple acceptance.

9 Reproducibility

From the repository root, install the dependencies in an isolated Python environment and run:

```
python solver.py --config config/2026-09.json --department imagiologia \
  --month 2026-09 --output-dir reports/2026-09 --time-limit 300
python evaluate.py --config config/2026-09.json \
  --schedule reports/2026-09/schedule.csv --output-dir reports/2026-09
python experiments.py --config config/2026-09.json --month 2026-09 \
  --output-dir experiments/2026-09
python robustness.py --config config/2026-09.json \
  --schedule reports/2026-09/schedule.csv --output-dir robustness/2026-09
python explanations.py
python -m unittest discover
```

Exact command-line flags are documented by each program's `-help` output. The generated JSON reports should be retained with the submitted PDF because they contain the machine-readable evidence behind the tables.

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A Constraint Catalogue

Identifier	Meaning
one_shift_per_day	At most one shift per worker and calendar day.
staffing_minimum	Afternoon and night minima are mandatory; weekday mornings have a hard floor of six and a preferred target of 10.
availability	Vacations, consultation assignment, sick leave, and shift eligibility are respected.
weekly_hours	Weekly assigned time does not exceed the configured legal limit.
monthly_hours	Monthly hours respect contractual capacity and carried rest, with bounded overtime where enabled.
afternoon_rest	A standard afternoon cannot be followed by a morning.
early_afternoon_night	An early afternoon cannot be followed by a night.
night_rest	A night cannot be followed by work the next day.
night_spacing	Nights assigned to one worker have the required spacing.
night_fairness_band	Eligible workers receive floor/ceiling shares of nights.

B Objective Weights in the Main Run

Component	Weight
Workday balance	100,000
Workload imbalance	100
Excess work streak	500
Isolated work day	300
Isolated rest day	200
Excess afternoon streak	400
Night ramp mismatch	150
Night after vacation	1,000
Holiday rotation	1,000
Special-holiday rotation	2,000
Weekend imbalance	30
Consecutive weekends	20
Split weekend	10
Vacation-adjacent weekend	5
Morning shortfall (Mon/Wed/Fri)	10,000,000
Morning shortfall (Tue/Thu)	500

Component	Weight
Monthly overtime	50